Exact Computer-Assisted Exclusion of the $(72, 108)$ Frontier in the Two-Dimensional Jacobian Problem

With context from the newly announced counterexample in dimension three

Billel Helali
Computational verification assisted by OpenAI Codex

21 July 2026

Status and scope. The identities reported below have been replayed in exact characteristic-zero arithmetic for the transcribed Laurent/Newton systems. The exclusion of the degree pair $(72, 108)$, and hence the degree-125 lower bound, remains conditional on the correctness and exhaustiveness of the published reduction of Guccione–Guccione–Horruitiner–Valqui and on its faithful transcription. This note does *not* prove the two-dimensional Jacobian conjecture. External mathematical and computational review is required before the result should be treated as established.

Abstract

An explicit noninjective Keller map in three variables was publicly announced in July 2026. A direct exact calculation gives constant Jacobian determinant -2 and exhibits three distinct points with a common image; adjoining identity coordinates therefore gives counterexamples in every dimension $n \geq 3$. The plane case remains separate. For the last degree pair $(72, 108)$ left below 125 by the Laurent/Newton analysis of Guccione, Guccione, Horruitiner and Valqui, we reconstruct and independently replay exact characteristic-zero ideal certificates for both residual configurations. The hard certificate is an 89.1 MB identity over a degree-five number field. It is obtained from a fixed 1925×1925 rational subsystem selected modulo 71 and then verified against all 2010 scalar equations. An independent `gmpy2` verifier checks 13,410 coefficient-field products, or 335,250 scalar products. Subject to the published reduction and its transcription, the pair $(72, 108)$ is impossible and any planar counterexample must have maximum degree at least 125.

Contents

1 Introduction: the dimensional split **2**

2 The announced counterexample in dimension three **2**

3 The planar $(72, 108)$ frontier **3**

4 Exact Laurent reduction **3**

5 Certificate architecture **3**

5.1 Case 2 4

5.2 Case 1: exhaustive branching and the specialization $h = 0$ 4

5.3 Case 1: the hard identity 4

6 Construction of the hard certificate **4**

7 Independent replay and reproducibility **5**

8 Main result and limitations **5**

9 Conclusion **6**

1 Introduction: the dimensional split

For a polynomial map $F = (F_1, \dots, F_n) : \mathbb{C}^n \to \mathbb{C}^n$, the classical Jacobian conjecture asserts that a nonzero constant determinant $\det JF$ forces F to have a polynomial inverse. The conjecture was introduced by Keller in 1939 [1].

The situation changed sharply on 20 July 2026, when Levent Alpöge circulated an explicit three-dimensional polynomial map on X [3]; an expository note by Zhang records the formula and its immediate consequences [4]. Section 2 gives a self-contained exact check. The map is a noninjective Keller map, so JC(3) is false; adjoining untouched coordinates proves the failure of JC(n) for every $n \geq 3$.

This development does not decide JC(2). The plane problem is now the remaining positive-dimensional case of the classical conjecture. The present work concerns only its known $(72, 108)$ degree frontier. Guccione, Guccione, Horruitiner and Valqui showed that every hypothetical plane counterexample of maximum degree below 125 must have degree pair $(72, 108)$, up to order [2]. Their Proposition 4.3 reduces this pair to two explicit Laurent/Newton configurations. Our contribution is the reconstruction and exact replay of algebraic certificates excluding the coefficient systems transcribed from both configurations.

2 The announced counterexample in dimension three

Consider $F = (P, Q, R) : \mathbb{C}^3 \to \mathbb{C}^3$ with

$$\begin{aligned} P &= (1+xy)^3 z + y^2(1+xy)(4+3xy), \\ Q &= y + 3x(1+xy)^2 z + 3xy^2(4+3xy), \\ R &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

Its component degrees are $(7, 6, 4)$. Exact symbolic expansion gives

$$\det\left(\frac{\partial(P,Q,R)}{\partial(x,y,z)}\right) = -2.$$

Moreover,

$$\begin{aligned} F\left(0, 0, -\tfrac{1}{4}\right) = F\left(1, -\tfrac{3}{2}, \tfrac{13}{2}\right) = F\left(-1, \tfrac{3}{2}, \tfrac{13}{2}\right) \\ = \left(-\tfrac{1}{4}, 0, 0\right). \end{aligned}$$

The three source points are distinct. Thus F has nonzero constant Jacobian but is not injective, and hence cannot be a polynomial automorphism. Multiplying one output coordinate by $-1/2$ normalizes the determinant to 1. Finally,

$$F_n(x_1, x_2, x_3, x_4, \dots, x_n) = (F(x_1, x_2, x_3), x_4, \dots, x_n)$$

extends the example to every $n \geq 3$.

Remark 2.1. The verification above is elementary and exact; it does not depend on numerical evidence. The historical attribution in this note follows the public announcement [3] and the contemporaneous record [4]. As of the date of this manuscript, the announcement is extremely recent and should be cited as such.

3 The planar $(72, 108)$ frontier

The main published input is the following consequence of Theorem 2.1 and Proposition 4.3 of [2]: below maximum degree 125, only $(72, 108)$ and its symmetric pair remain, and a counterexample of that degree type would yield Laurent polynomials $P, Q \in K[x, x^{-1}, y]$ over an algebraically closed characteristic-zero field satisfying

$$[P, Q] = x^2$$

in one of the following two Newton configurations:

$$\begin{aligned} \text{Case 1:} \quad & N(P) = \operatorname{conv}\{(0,0),(1,0),(8,14),(8,16),(0,8)\}, \\ & N(Q) = \operatorname{conv}\{(0,0),(2,1),(12,21),(12,24),(0,12)\}; \\ \text{Case 2:} \quad & N(P) = \operatorname{conv}\{(0,0),(1,0),(8,14),(8,16)\}, \\ & N(Q) = \operatorname{conv}\{(0,0),(2,1),(12,21),(12,24)\}. \end{aligned}$$

We emphasize the logical boundary: our computation proves identities for the systems obtained from these two polygonal cases. Promoting those identities to a theorem about all plane maps requires the reduction in [2] to be exhaustive and the local transcription and normalization to be exact.

4 Exact Laurent reduction

Set $t = xy^2$ and $z = y^{-1}$. Then $[t, z]_{x,y} = -1$ and $x^2 = t^2 z^4$. The relevant upper bands can be written

$$P = A(t)z^2 + B(t)z + C(t), \qquad Q = D(t)z^3 + E(t)z^2 + F(t)z + G(t).$$

Equating coefficients in $[P, Q] = x^2$ gives

$$2AD' - 3A'D = t^2, \tag{J4}$$
$$2(AE' - A'E) + (BD' - 3B'D) = 0, \tag{J3}$$
$$(2AF' - A'F) + (BE' - 2B'E) - 3C'D = 0, \tag{J2}$$
$$2AG' + (BF' - B'F) - 2C'E = 0, \tag{J1}$$
$$BG' - C'F = 0. \tag{J0}$$

The replay checks the coordinate bracket, the identity $x^2 = t^2 z^4$, all five formulas, and the polygonal lattice supports exactly.

With normalized nonzero vertex coefficients,

$$A = t + a_2 t^2 + \cdots + a_7 t^7 + t^8, \qquad D = d_2 t^2 + \cdots + d_{12} t^{12}.$$

Equation (J4) solves the eleven coefficients of D and leaves six compatibility equations in $a_2, \ldots, a_7$. Exact rational Gröbner and FGLM calculations yield an irreducible degree-35 polynomial $H(a_7)$ together with five relations linear in $a_2, \ldots, a_6$. Subsequent residual systems descend to

$$L = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26).$$

5 Certificate architecture

Two elementary observations carry the logical proof. If $1 = \sum_i T_i E_i$, then the E_i have no common zero. If $h = \sum_i T_i E_i$, then every common zero satisfies $h = 0$. All displayed certificates are identities in polynomial rings over exact number fields.

5.1 Case 2

Exact elimination in equations (J3)–(J0) leaves four residual generators $R_1, \dots, R_4$ over a degree-35 field. A Macaulay system of degree 8, with dimensions 165×182, produces exact multipliers satisfying

$$1 = T_1R_1 + T_2R_2 + T_3R_3 + T_4R_4.$$

The serialized certificate is evaluated again over the number field; the exact remainder is 1. Hence Case 2 has no solution.

5.2 Case 1: exhaustive branching and the specialization $h = 0$

The negative z-bands produce thirteen compatibility equations. Their first factorization forces the exhaustive split $s = c$ or $s = -c$, where $c \neq 0$. In each branch a checked relation

$$E_6 - \lambda E_2 = S\Lambda^2, \qquad S \neq 0,$$

forces an affine-linear form Λ to vanish, eliminating one variable without division by a variable expression.

For $h = 0$, exact unit-ideal certificates are reconstructed separately for the two branches using the seven pre-division equations:

$$1 = \sum_i A_i E_i\big|_{h=0}.$$

Thus neither branch has a solution with $h = 0$. The use of pre-division equations is important: it avoids losing a component through an unrecorded saturation or variable division.

5.3 Case 1: the hard identity

For the branch $s = c$, four reduced generators in $L[h, u_1, u_2]$ admit the identity

$$\boxed{h = T_1E_1 + T_2E_2 + T_3E_3 + T_4E_4.} \tag{1}$$

Therefore every common zero would satisfy $h = 0$, which the preceding unit certificate excludes. The involution

$$(h, u_1, u_2) \longmapsto (h, -u_1, -u_2)$$

transports the equations and identity (1) exactly to $s = -c$. Both branches, and therefore Case 1, are empty.

6 Construction of the hard certificate

The multiplier ansatz for (1) has 385 coefficients in L and gives 402 coefficient equations in L. Expanding on the basis $1, w, \dots, w^4$ produces a rational linear system with

$$2010 \text{ equations and } 1925 \text{ unknowns.}$$

A deterministic set of 1925 scalar rows was selected by exact elimination modulo 71. The square minor B_{71} satisfies

$$\det(B_{71}) \equiv 10 \pmod{71},$$

so the chosen support is nonsingular modulo 71 and hence nonsingular over $\mathbb{Q}$. The corresponding 1925×1925 system was rebuilt and solved in exact rational arithmetic using FLINT. The resulting vector has 1925 nonzero rational coordinates; its largest numerator and denominator have 85,342 and 85,275 bits, respectively. After serialization, the human-readable identity occupies 89,105,967 bytes.

The selected square subsystem is only the construction mechanism. Soundness comes from substituting the recovered multipliers into all 402 equations over L, equivalently all 2010 rational scalar rows, and checking a zero residual everywhere.

7 Independent replay and reproducibility

The full replay is launched from the exact workspace by

```
./verify_all.sh
```

and terminates with

```
ALL_SERIALIZED_EXACT_CERTIFICATES_PASS
GMPY2_EXACT_PASS
SYSTEM_SYMMETRY_PASS
BRANCH1_EXACT_IDENTITY_PASS
BRANCH2_EXACT_IDENTITY_PASS
JC2_72_108_EXACT_REPLAY_PASS
```

The independent `gmpy2` path parses the large text certificate, normalizes its 1925 rational coordinates, and evaluates the identity without relying on the FLINT solver that produced it. It performs 13,410 coefficient-field products, corresponding to 335,250 scalar products. The branch symmetry is also evaluated as a polynomial identity rather than inferred from matching dimensions.

Table 1: Core reconstructed artifacts. SHA-256 digests identify the exact files used in this replay.

Artifact	SHA-256
`hard/h_certificate_exact.txt`	`0e48ffab32469ef8405a6945b16cf1521ddeb3c592ae4e5051968110a4dc656a`
`case2_exact_certificate.pkl`	`cfbc3c39d7a28013671144f43ef76f0498542eaf6d562dd624bba3311194e4aa`
`h0_branch1_exact_certificate.pkl`	`664de005e99bc6a0e61ba479ba64ed57ddbfa9c5399b955cbb12237ac70f8186`
`h0_exact_certificate.pkl`	`d7844c2f8edea62be4e3a7fe8a160dc4b7b70efd1262993ec6d2ed7e78722a1a`
`verify_all.sh`	`f40bacfa84d915b375b4158109d5120b2d964a8d8012be770750d310abfb5837`

The reconstructed objects are mathematically equivalent to the historically described certificates but not byte-for-byte copies; serialization and textual formatting changed, so their hashes necessarily differ. The manifest `RECONSTRUCTED_CERTIFICATES.sha256` is the authority for this replay.

8 Main result and limitations

Theorem 8.1 (Exact exclusion of the transcribed systems)**.** *The two characteristic-zero coefficient systems transcribed from Cases 1 and 2 of the* $(72, 108)$ *Laurent/Newton reduction have no solutions.*

Proof. Case 2 is inconsistent because its four residual generators admit an exact unit certificate. In Case 1 the split $s = \pm c$ is exhaustive. On each branch, the hard identity forces $h = 0$, while the corresponding pre-division unit certificate excludes $h = 0$. The exact involution verifies the transport from the first branch to the second. □

Corollary 8.2 (Conditional degree bound)**.** *Assume that Proposition 4.3 of [2] is exhaustive for the* $(72, 108)$ *configuration and that the local normalization and transcription are faithful. Then* $(72, 108)$ *cannot be the degree pair of a plane Jacobian counterexample. Consequently, every hypothetical counterexample to* JC(2) *has*

$$\max\{\deg P, \deg Q\} \geq 125.$$

This conclusion does not prove JC(2): configurations of maximum degree at least 125 remain outside its scope. It also does not replace a line-by-line external audit of the passage from the published Newton polygons to the implemented coefficient systems. The appropriate public claim is therefore an *exact computer-assisted candidate exclusion*, conditional on that theoretical interface, rather than a completed proof of the plane conjecture.

9 Conclusion

The newly announced three-dimensional example creates a clear dimensional divide: the classical Jacobian conjecture is false in every dimension $n \geq 3$, while dimension two remains unresolved. Against that background, the exact reconstruction reported here removes, conditionally on the published reduction, the sole degree pair left below 125. The computational content is finite and replayable: serialized unit certificates, a large exact membership identity, fixed hashes, an independent arithmetic verifier, and a single terminal success marker. The next necessary step is independent specialist review of both the theoretical transcription and the certificate replay.

Data and code availability

The LaTeX source, exact certificates, reconstruction programs, manifests, and replay scripts are distributed with the accompanying research archive. The principal certificate is too large to print but is directly machine-verifiable from `hard/h_certificate_exact.txt`.

Acknowledgment of computational assistance

OpenAI Codex assisted with certificate reconstruction, verification scripting, and preparation of this manuscript. All mathematical claims in the note are tied to exact machine-checkable identities or to explicitly cited published reductions; responsibility for validation and interpretation remains with the author.

References

[1] O.-H. Keller, *Ganze Cremona-Transformationen*, Monatshefte für Mathematik und Physik **47** (1939), 299–306.

[2] J. A. Guccione, J. J. Guccione, R. Horruitiner, and C. Valqui, *Increasing the degree of a possible counterexample to the Jacobian Conjecture from 100 to 108*, arXiv:2204.14178 (2022), https://arxiv.org/abs/2204.14178.

[3] L. Alpöge, *Public announcement of an explicit three-dimensional Keller map*, X post, 20 July 2026, https://x.com/__alpoge__/status/2079028340955197566.

[4] Z. Zhang, *Direct Consequences of the Three-Dimensional Counterexample to the Jacobian Conjecture*, expository note, 20 July 2026, https://zzhang-iu.github.io/papers/direct-consequences-jacobian/.